

Overview of Machine Learning and Predictive Modeling

What is ML:

Machine learning is a field of data science/statistics/artificial intelligence which involves creating **models** to enable computers to “learn” patterns in data and make predictions based on these patterns

- A model is a function which can map input data to output data:
 $f(\text{input}) = \text{output}$

Example: create a model to predict real estate prices

- Input data: square footage, location, age, number of bedrooms, etc.
- Output data: the price (a continuous value)

Example: predict whether a patient has cancer

- Input data: age, gender, BMI, family history, environmental factors, etc.
- Output data: does/doesn't have cancer (a discrete label)

Overview of ML

Some ML terms:

Dataset: a collection of inputs/outputs

Samples: entries in the dataset
(rows in the dataset)

Features: the input data (usually X , a 2D array (N_{samp}, N_{feat}))
(columns in the dataset)
(sometimes called predictors)

Labels: the output data (usually y , a 1D array (N_{samp}))
(sometimes called “target”)
(a special column in the dataset)

Model: a function f which predicts labels based on features

Training: the process by which the model “learns” the patterns in a dataset

Testing: the process by which a model’s performance/accuracy is evaluated

Loss function: a function which evaluates the amount of error in the model
(sometimes called error, must be minimized)
(training is minimizing the loss function)

A Taxonomy of ML (pt 1):

One way to classify the different types of ML is based on the nature of the task (i.e., the features and labels):

- Supervised learning: a type of learning where the dataset has features AND labels and the goal is to predict the labels based on the features
 - Regression (label is continuous)
 - Classification (label is categorical/discrete)
- Unsupervised learning: a type of learning where the dataset ONLY has features and the goal is to assign a set of reasonable labels
 - Clustering
- Generative learning: a type of learning that models the distribution of the data so that similar data can be generated (terms features/labels aren't used as much)
 - AI chatbots
 - AI image generators

A Taxonomy of ML (pt 2):

Another way to classify ML methods is based on the nature of the algorithms used:

- Linear models: model the relationship between features/labels using linear equations (e.g., linear regression and logistic regression)
- Instance-based learning: makes predictions based on the similarity of new instances to existing data points (e.g., k -nearest neighbors)
- Tree-based methods: use decision trees as the fundamental building block (e.g., decision trees, random forests, and gradient boosting)
- Ensemble methods: combine multiple models to improve overall performance (e.g., bagging and boosting)
- Deep learning: uses neural networks with multiple layers (e.g., dense NN, convolutional NN, recurrent NN, LSTM's, transformers)

Overview of ML

Python libraries for data processing:

- NumPy, SciPy
- Pandas

Python libraries for visualization:

- Matplotlib, Seaborn
- Plotly, Bokeh

Python libraries for ML:

- Scikit-learn
- Statsmodels
- Tensorflow/Keras, PyTorch (neural networks)
- XGBoost, LightGBM (gradient boosting)

Garbage in, garbage out:

No matter how clever and advanced a model is, if it is given bad data...

- useless features that don't relate to the labels
- not enough features
- missing values, errors in data
- insufficient number of samples
- too little variety (not all possible cases are covered)
- trying to predict beyond the scope of the data (extrapolating)

then it will make poor predictions

How to ML:

So, what is the correct way to perform ML and (more broadly) data science in general and avoid these and other pitfalls?

CRISP-DM: CRoss-Industry Standard Process for Data Mining

- a widely-used methodology for data mining and analytics projects
- provides a structured approach to guide the entire DS/ML process
- an iterative 6-step process, allowing for revisiting and refining at each step